Q.1) Write a code to Read a file and append lines to a list.

```
lines_list = []
with open("try.txt", "r") as file:   # Replace 'sample.txt' with your filename
    for line in file:
        lines_list.append(line.strip())  # strip() removes newline characters

print(lines_list)
```

```
E:\vita_CDAC\3_PYTHON\.venv\Scripts\python.exe E:\vita_CDAC\3_PYTHON\placementprep\day3test.py
['try:', 'mydatabase=m.connect(host="localhost",user="root",password="root",database="pythondb1")', 'cursor=my

Process finished with exit code 0
```

Q.2) Write a code to catch an Exception in python?

```
from exceptiongroup import catch

try:
    num = int(input("Enter a number: "))
    print(num)
except ValueError as e:
    print("invlaid input")
```

```
E:\vita_CDAC\3_PYTHON\.venv\Scripts\python.exe E:\vita_CDAC\3_PYTHON\placementprep\day3test.py
Enter a number: 10
10

Process finished with exit code 0
```

Q.3) Write a Python function that accepts a list containing strings and integers.
Merge all string elements using # and add all integer elements.
e.g.
input list is
['100', 'welcome', 'hi', '200', '300', 'bye', 'welldone', '500']
Output should be:
welcome#hi#bye#welldone#
1100

```
def process_list(input_list):
   strings = []
   total = 0

   for item in input_list:
      if item.isdigit():  # Check if the element is a number (string form)
         total += int(item)
      else:
         strings.append(item)

   merged_strings = "#".join(strings) + "#"  # Add trailing #
   return merged_strings, total


# Example usage
input_list = ['100', 'welcome', 'hi', '200', '300', 'bye', 'welldone', '500']
result = process_list(input_list)
print(result[0])  # Output: welcome#hi#bye#welldone#
print(result[1])  # Output: 1100
```

```
E:\vita_CDAC\3_PYTHON\.venv\Scripts\python.exe E:\vita_CDAC\3_PYTHON\placementprep\day3test.py
welcome#hi#bye#welldone#
1100

Process finished with exit code 0
```

Q.4) Write a script to sort a dictionary based on its values and find the sum of middle two values
input_dict = {"x": 5, "y": 15, "z": 25}
Output:
Sorted Dictionary: {'x': 5, 'y': 15, 'z': 25}
Sum of middle two values: 15 + 5 = 20
or
input_dict = {"x": 5, "y": 15, "z": 25,"p":12}
Output:
Sorted Dictionary: {'x': 5, 'p': 12,'y': 15, 'z': 25}
Sum of middle two values: 12 + 15 = 27

```
def sort_dict_and_sum_middle(input_dict):
   # Sort dictionary by values
   sorted_dict = dict(sorted(input_dict.items(), key=lambda x: x[1]))
```

```
    print("Sorted Dictionary:", sorted_dict)

    # Extract values
    values = list(sorted_dict.values())

    # Find middle two values
    n = len(values)
    if n % 2 == 0:  # Even number of elements
        middle_sum = values[n // 2 - 1] + values[n // 2]
    else:  # Odd number of elements
        middle_sum = values[n // 2 - 1] + values[n // 2]  # As per given examples

    print("Sum of middle two values:", middle_sum)


# Example usage
sort_dict_and_sum_middle({"x": 5, "y": 15, "z": 25})
sort_dict_and_sum_middle({"x": 5, "y": 15, "z": 25, "p": 12})
```

```
E:\vita_CDAC\3_PYTHON\.venv\Scripts\python.exe E:\vita_CDAC\3_PYTHON\placementprep\day3test.py
Sorted Dictionary: {'x': 5, 'y': 15, 'z': 25}
Sum of middle two values: 20
Sorted Dictionary: {'x': 5, 'p': 12, 'y': 15, 'z': 25}
Sum of middle two values: 27

Process finished with exit code 0
```